

AP Statistics Quiz Study Guide

Lessons 1.1–1.3: Categorical Data Foundations

Statistical studies → categorical tables → graphs and association

- Built from:** Lesson 1.1, 1.2, and 1.3 notes, homework, and answer keys
- Best use tonight:** Read the concept pages, cover the answers, complete the practice set, then grade yourself from the key.
- Quiz goal:** Name the correct denominator, compare conditional percentages, and explain conclusions in context.

The complete quiz checklist

By the end, you should be able to do each task without notes.

- ✓ Identify individuals and variables.
- ✓ Classify variables as categorical or quantitative.
- ✓ Distinguish population, sample, and census.
- ✓ Distinguish prospective and retrospective studies.
- ✓ State a realistic limitation; avoid causal claims.
- ✓ Find frequency and relative frequency.
- ✓ Read marginal, joint, and conditional values.
- ✓ Choose explanatory and response variables.
- ✓ Decide whether two categorical variables are associated.
- ✓ Read side-by-side and segmented bar graphs.
- ✓ Explain the height, width, and area in a mosaic plot.
- ✓ Turn percentages back into counts, and counts back into a total.
- ✓ Say which comparisons a graph does and does not support.
- ✓ Identify misleading scales and misleading pictures.

A focused 60-minute plan

- 0–15 min** Memorize the vocabulary and denominator rules in the concept section.
- 15–30 min** Redo the two class-data worked examples without looking at the blue solutions.
- 30–50 min** Complete the mixed practice sets without using the answer key.
- 50–60 min** Grade with the answer key. Redo only the questions you missed.

Lesson 1.1: Statistical Studies

What a statistical study does

A statistical study collects data from a **sample** to answer an **investigative question** about a **population**. A conclusion is strongest when the sample represents the population well.

Term	Meaning	Question to ask
Individual	One object or person described by the data	“Who or what is each row about?”
Variable	A characteristic recorded for each individual	“What changes from individual to individual?”
Population, N	The entire group the conclusion is about	“Who do researchers want to understand?”
Sample, n	The subset from which data were actually collected	“Who was actually measured?”
Census	Data collected from every individual in the population	“Was everyone in that population included?”

Wording a variable correctly

A variable must describe the recorded value for *each individual*. Write “whether or not an adult owns a pet,” not merely “owning a pet.” Write “reaction time in milliseconds,” not merely “reaction.”

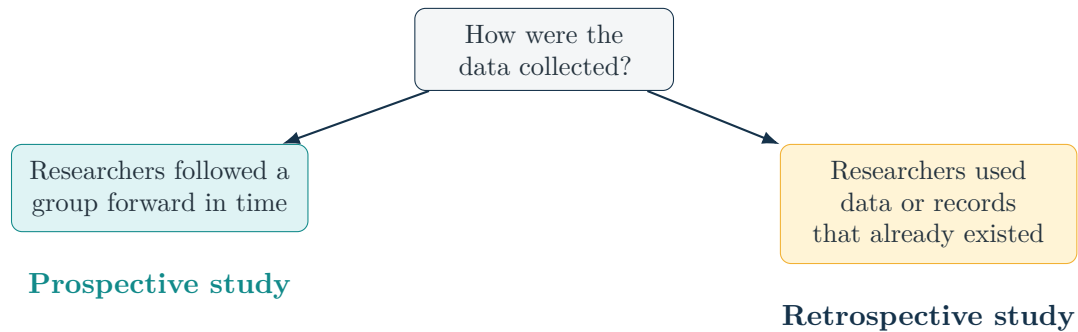
Categorical vs. quantitative variables

Type	Definition	Examples from the lessons
Categorical	Values are category names or group labels. Arithmetic on the labels is not meaningful.	Drinks coffee daily (yes/no); long or short walk; developed disease (yes/no); preferred class
Quantitative	Values are numerical measurements or counts. Arithmetic is meaningful.	Reaction time; stress score; number of steps; exam score; hours worked

Numbers can still be categorical

A jersey number, ZIP code, or student ID is a label, so it is categorical. Ask whether averaging the values would make sense.

Prospective vs. retrospective



Evidence words matter: “monitored for 8 years” supports prospective; “examined existing survey data” supports retrospective.

Limitations and conclusions

Common valid limitations in the provided materials include:

- a short measurement period may not represent usual behavior;
- self-reported responses may be inaccurate;
- the sample may not represent the target population;
- another variable may explain the association (confounding);
- an observational study shows association, not cause and effect.

Safe conclusion language

Say “was associated with,” “tended to,” or “was more likely.” Do not say one variable *caused* the other unless a well-designed experiment supports that claim.

Lesson 1.2: Analyzing Categorical Data

One categorical variable

Display	What it shows	Key rule
Bar graph	Frequency or relative frequency for categories	Bars are separated; a frequency axis must start at 0.
Pie chart	Each category's share of the whole	All slices total 100%.

Frequency language — and how to run it backwards

$$\text{relative frequency of a category} = \frac{\text{frequency in that category}}{\text{total number of individuals}} \quad \text{percent} = 100\%(\text{relative frequency})$$

Solve for whichever piece is missing:

$$\text{count} = (\text{relative frequency})(\text{total}), \quad \text{total} = \frac{\text{count}}{\text{relative frequency}}.$$

From the notes: 9 students chose Music and Music was 30% of the sample, so the total sampled is $9/0.30 = 30$ students; a missing category count is then $30 - (6 + 9 + 4 + 4) = 7$.

Name the individuals first

“The 30 major league baseball teams” are the individuals; league (categorical), stadium capacity (quantitative), wins last season (quantitative), and whether or not the team made the playoffs (categorical) are the variables. Individuals are not always people.

Two-way tables: choose the denominator first

Suppose the rows are categories of variable A and the columns are categories of variable B .

	B_1	B_2	Total
A_1	a	b	$a + b$
A_2	c	d	$c + d$
Total	$a + c$	$b + d$	n

Type	Meaning	Denominator
Marginal	One variable only; value found in a margin	Grand total n
Joint	Two conditions connected by “and”	Grand total n
Conditional	One condition restricted by “of <i>those who...</i> ,” “among,” or “given”	Total of the given group

Fast denominator test

$$P(A_1) = \frac{a+b}{n} \quad \text{marginal}$$

$$P(A_1 \text{ and } B_1) = \frac{a}{n} \quad \text{joint}$$

$$P(A_1 | B_1) = \frac{a}{a+c} \quad \text{conditional: denominator is all of } B_1$$

The condition after “given” or “of those who” determines the denominator. Watch out: “of *all* the students surveyed” is not a condition — it just names the grand total.

Worked example from the class notes

Favorite elective	Math	English	Total
Art	2	4	6
Music	5	4	9
P.E.	4	3	7
World Language	1	3	4
Technology	4	0	4
Total	16	14	30

Marginal: percent of all students who chose P.E.

Joint: percent who chose Math *and* Art

$$\frac{7}{30} = 0.233 = 23.3\%$$

$$\frac{2}{30} = 0.067 = 6.7\%$$

Conditional: percent who chose Technology, *given Math*

Reverse conditional: percent who chose Math, *given Technology*

$$\frac{4}{16} = 0.25 = 25\%$$

$$\frac{4}{4} = 1 = 100\%$$

Reversing the condition

$P(\text{Technology} | \text{Math})$ and $P(\text{Math} | \text{Technology})$ use different denominators. Read the wording before calculating.

The homework table you must be able to do cold

This CBS News table (age vs. preferred listening format) appears in both Lesson 1.2 homework problems. Fill in every margin *before* answering anything.

Preferred format	Age < 45	Age 45+	Total
Streaming	386	306	692
Radio	138	389	527
Digital files	124	109	233
CDs	29	95	124
Total	677	899	1576

Type	The question, word for word	Answer
Joint	“What percent of <i>all</i> adults surveyed are 45 or older <i>and</i> prefer CDs?”	$\frac{95}{1576} = 0.060 = 6.0\%$
Marginal	“What percent of <i>all</i> adults surveyed prefer listening to the radio?”	$\frac{138 + 389}{1576} = \frac{527}{1576} = 33.4\%$
Conditional	“ <i>Of those adults that prefer streaming</i> , what percent are less than 45?”	$\frac{386}{692} = 0.558 = 55.8\%$
Conditional	“ <i>Of the adults less than 45</i> , what percent prefer streaming?”	$\frac{386}{677} = 0.570 = 57.0\%$
Conditional	“ <i>Of the adults 45 or older</i> , what percent prefer streaming?”	$\frac{306}{899} = 0.340 = 34.0\%$

The same 386, three different denominators

386/1576 (joint), 386/692 (given streaming), and 386/677 (given age < 45) are three different answers to three different questions. Find the phrase that names the group *before* you divide: “of all” $\Rightarrow$ 1576; “of those who prefer streaming” $\Rightarrow$ 692; “of the adults under 45” $\Rightarrow$ 677.

Association from this table: 57.0% of adults under 45 prefer streaming, compared with only 34.0% of adults 45 or older. The conditional distributions differ, so age and preferred listening format *are* associated: knowing a person’s age helps predict their preferred format.

A numerator can span two cells

“For colleges in Texas, what proportion are private?” with a Public / Private-nonprofit / Private-for-profit table means *add both private columns* and divide by the Texas row total: $\frac{71+84}{262} = \frac{155}{262}$. Not $\frac{155}{996}$ — the question restricts to Texas.

Association and Graphs for Two Categorical Variables

Explanatory and response variables

The **explanatory variable** may help explain or predict the **response variable**. In the class example, favorite core class can be treated as explanatory and favorite elective as response. This choice provides direction for the conditional distributions you compare.

How to decide whether there is an association

Two categorical variables are associated when knowing the value of one helps predict the value of the other. Operationally:

1. Choose the explanatory groups.
2. Calculate the response distribution *within each group*.
3. Compare corresponding conditional percentages.
4. If the distributions differ meaningfully, state that there is an association and describe a specific difference in context.

For the class table, 25% of Math-preferring students chose Technology, while 0% of English-preferring students did. The conditional distributions differ, so core preference and elective preference are associated in this sample.

Strong sentence frame

There **is** an association between **core-class preference** and **elective preference** because the conditional distribution of **elective preference** differs between the core-class groups. For example, **25% of Math-preferring students chose Technology, compared with 0% of English-preferring students.**

No association

If there is no association, corresponding segments have the same relative frequencies in every group. The raw counts may still differ when group sizes differ.

Lesson 1.3 displays

Display		What stays visible	What to compare
Side-by-side graph	bar	Separate bars for each response category within each explanatory group	Corresponding bar heights (relative frequencies)
Segmented graph	bar	Each group is one equal-width, 100%-tall bar	Corresponding segment heights
Mosaic plot		Segment height is conditional percent; bar width is proportional to group size	Heights for association; widths for sample sizes; areas for counts

Mosaic plot meanings

height = conditional proportion,
width = group's share of the sample,
area = joint share of the full sample.

If a group is twice as large, its mosaic bar is twice as wide.

Worked mosaic example from the class notes

Teachers and students voted for a mascot. There were 100 teachers and 500 students.

Percentages (what the graph shows)	Rams	Falcons	Prairie Dogs	Group size
Teachers	80%	10%	10%	100
Students	30%	60%	10%	500

Convert each percentage to a count using

$$\text{count} = (\text{conditional proportion})(\text{group size}).$$

Counts	Rams	Falcons	Prairie Dogs	Total
Teachers	$0.80(100) = 80$	$0.10(100) = 10$	$0.10(100) = 10$	100
Students	$0.30(500) = 150$	$0.60(500) = 300$	$0.10(500) = 50$	500
Total	230	310	60	600

If you look only at the two percentage bars, Rams (80%) look dominant because the teacher bar is drawn the same width as the student bar. Once the group sizes are used, **Falcons win the vote, 310 to 230**. The student bar should be **five times as wide** as the teacher bar, and the Falcon rectangle then has the greatest total area.

Total count across groups (Lesson 1.3 homework #3)

When a mosaic plot gives conditional percentages and you know each group size, the overall count is a weighted sum:

$$\text{total} = \sum_{\text{groups}} (\text{conditional proportion})(\text{group size}).$$

Music class ($n = 30$) had 50% A's; the two no-music classes ($n = 60$) had 30% A's, so the number of A's overall is

$$(0.50)(30) + (0.30)(60) = 15 + 18 = 33 \text{ students.}$$

For a single group, just multiply once: 10% of the 60 no-music students earned an E, so $(0.10)(60) = 6$ students.

Percent vs. number

A higher percentage does not guarantee a higher count unless the group sizes are known. In the breakfast homework, the grades 8–12 group had the higher percentage that “sometimes skips” ($\approx 19\%$ vs. $\approx 16\%$), but the number of students sampled from each grade group was never given, so you *cannot* say which group had more breakfast-skippers. The correct answer is “we don’t know for certain,” with that reason stated.

“Which statement MUST be true?” — the two reading rules

1. **Segmented bar graph** (all bars equal width, all 100% tall): you may compare *within* one bar, because both segments share that group’s total. You may compare *percentages* across bars. You may **not** compare *counts* across bars *from the graph alone* — the group sizes are hidden. If the problem states the group sizes separately, multiply and compare away.
2. **Mosaic plot** (widths proportional to group size): everything above, plus you may compare *areas* across bars, because area is proportional to count.

So “for high school C, more students took AP Statistics than AP Calculus” can be read off a segmented bar graph (same bar, same denominator), while “high school B and C enrolled the same number in AP Calculus” cannot. And “there are more 12th graders on football than on volleyball” needs a *mosaic* plot, where the football bar’s width carries the team size.

Misleading graphs

- A frequency bar graph whose vertical axis does not start at 0 exaggerates differences.
- A pictograph that changes both height and width makes the *area* grow too quickly.
- Comparing counts when only percentages are shown can be invalid if group sizes are unknown.
- Three-dimensional effects and inconsistent scales can distort comparisons.

Strong graph critique

The graph is misleading because the vertical axis begins above 0, which visually exaggerates the differences between the category frequencies. A corrected bar graph should start the frequency axis at 0 and keep equal scale intervals.

One-Page Memory Sheet

When you see...	Think...
“who was actually measured?”	sample
“who is the conclusion about?”	population
“every member”	census
“followed/monitored for years”	prospective
“used past records/examined existing data”	retrospective
category names or labels	categorical variable
measured or counted amount	quantitative variable
“of <i>all</i> ” with one category	marginal; divide by grand total
“of <i>all</i> ” with “and”	joint; divide by grand total
“of <i>those who...</i> ,” “among,” “given”	conditional; divide by that group’s total
same 100% composition across groups	no association
different conditional distributions	association
segmented bar height	conditional relative frequency
mosaic bar width	relative group size
mosaic rectangle area	joint relative frequency / share of all individuals
percentage but no group size	cannot automatically determine a count
percentage <i>and</i> group size	count = (proportion)(group size)
two segments in the <i>same</i> bar	safe to compare; same denominator
counts across <i>different</i> segmented bars	not readable from the graph alone
bar graph axis begins above 0	differences are visually exaggerated
picture changes in height and width	area exaggerates the difference
observational association	do not claim cause and effect

Four formulas to memorize

Relative frequency

$$\frac{\text{category count}}{\text{total count}}$$

Joint relative frequency

$$\frac{\text{one interior cell}}{\text{grand total}}$$

Conditional relative frequency

$$\frac{\text{one interior cell}}{\text{given-group total}}$$

Count from a percent

$$(\text{decimal proportion})(\text{group size})$$

Four last-minute checks

1. Name the variable as what is recorded for each individual.
2. Underline the group the question restricts to, then divide by *that* total.
3. Use percentages, not raw counts, to compare groups of different sizes.
4. Describe an association with a specific comparison in context.

Practice Set A: Lesson 1.1

Try these without the notes. Show work and write in complete sentences when asked to explain.

1. Prospective study and variables

Researchers want to study whether regular energy-drink use is related to sleep and school performance. They select 800 tenth-grade students from a school district. At the beginning of the semester, each student reports whether they regularly use energy drinks (yes/no) and average nightly sleep in hours. Researchers then record each student's final exam score at the end of the semester.

- Identify the population and sample. _____
- Identify and classify the three variables. _____
- Is the study prospective or retrospective? Cite evidence. _____
- State one limitation and explain why the study cannot establish cause and effect. _____

2. Past records

A researcher uses archived records from 2,400 students who graduated between 2020 and 2025. For each student, the researcher records whether the student took an art course and the student's graduation GPA.

- Is the study prospective or retrospective? Explain.
- Classify both variables.

3. Census or sample?

A coach records total minutes played and points scored for every player on the current varsity team. Which statement is correct?

- It is a sample because only athletes are included.
- It is an experiment because outcomes were recorded.
- It is a census of the current team's players.
- It must be retrospective because totals come from past games.

4. Writing variables precisely

Rewrite each vague phrase as a correctly worded variable for each individual, then classify it.

a. “phone use” _____

b. “transportation” _____

Practice Set B: Lesson 1.2

5. One categorical variable

A club records the primary way 130 members travel to school.

Mode	Car	Bus	Walk	Bicycle	Other
Frequency	52	38	25	10	5

a. Identify the individuals and the variable. Classify the variable.

b. Find the relative frequency for Bus and for Bicycle.

c. Name two appropriate displays for these data.

6. Two-way table mastery

A random sample of 240 students reports grade level and primary after-school activity.

	Sports	Arts	Job	None	Total
Sophomore	36	24	12	8	80
Junior	30	18	22	10	80
Senior	24	16	30	10	80
Total	90	58	64	28	240

For each, name the type (marginal, joint, or conditional), write the correct fraction, and calculate the percent.

a. Percent of all students whose primary activity is Arts.

b. Percent of all students who are Seniors and have a Job.

- c. Of the Seniors, the percent whose primary activity is a Job.
- d. Of the students with a Job, the percent who are Seniors.

7. Association in context

Treat grade level as the explanatory variable and after-school activity as the response.

- a. Compare the conditional percentages for Sports across the three grades.
- b. Compare the conditional percentages for Job across the three grades.
- c. Is there an association? Write a complete evidence-based sentence.

8. Misleading display

A bar graph shows approval rates of 94%, 96%, and 98%, but the vertical axis begins at 92%. Explain why the graph is misleading and describe a correction. _____

Practice Set C: Lesson 1.3

9. Mosaic plot without the picture

An exam study compares two study methods. The table shows counts.

	High score	Medium score	Low score	Total
App flashcards	20	12	8	40
Paper notes	24	40	16	80
Total	44	52	24	120

- a. Find the conditional score distribution for each study method.
- b. In a mosaic plot, how does the width of the Paper notes bar compare with the App flashcards bar? Explain.

- c. What percent of all students are in the App flashcards/High score rectangle? What does its area represent?
- d. Is study method associated with score category? Support your answer with a comparison.

10. Counts from a segmented bar graph

A segmented bar graph shows that 35% of Group A and 25% of Group B selected Option X.

- a. Can you determine which group has more people who selected X if group sizes are unknown? Explain.
- b. If Group A has 80 people and Group B has 160 people, find the count selecting X in each group. Which count is larger?

11. No association

In a survey, 60% of middle school students prefer Math and 40% prefer English. The survey finds no association between school level and preferred class. There are 100 middle school students and 200 high school students.

- a. What percentages of high school students prefer Math and English?
- b. Complete the counts.

	Math	English	Total
Middle school			100
High school			200
Total			300

- c. What would the corresponding segments look like in a segmented bar graph?

12. Multiple choice: what must be true?

A mosaic plot has two bars. The bar for School A is three times as wide as the bar for School B. In both bars, the “Yes” segment has height 40%. Which statement must be true?

- A) School A has three times as many “Yes” responses as School B.
- B) School B has a higher conditional percentage of “Yes.”
- C) There is an association between school and response.
- D) The two schools have the same number of “Yes” responses.

13. A numerator that spans two columns

A survey classified 400 restaurants in three cities by type.

	Fast food	Casual dining	Fine dining	Total
City A	60	45	15	120
City B	90	50	20	160
City C	70	40	10	120
Total	220	135	45	400

For restaurants in City B, what proportion are sit-down restaurants (casual dining or fine dining)?

- A) $\frac{70}{400}$
- B) $\frac{70}{160}$
- C) $\frac{70}{180}$
- D) $\frac{160}{400}$

14. Reading a segmented bar graph

A segmented bar graph (three equal-width bars, each 100% tall) shows after-school job status at three schools.

	Has a job	Does not have a job
School P	45%	55%
School Q	30%	70%
School R	45%	55%

Which statement *must* be true?

- A) Schools P and R have the same number of students with jobs.
- B) At School Q, more students do not have a job than have a job.
- C) School P has more students with jobs than School Q.
- D) There is no association between school and job status.

15. Mosaic plot with counts

A mosaic plot shows the percent of students who passed a fitness test in three PE periods. Period 1 has 20 students, Period 2 has 25, and Period 3 has 35. The passing percentages are 60%, 40%, and 80%.

- Which bar is widest, and why?
- How many students passed in all?
- Is there an association between period and passing? Support your answer.
- What does the area of the Period 3 / passed rectangle represent, as a percent of all students?

16. Running relative frequency backwards

A class recorded each student's favorite lunch: Pizza 12, Sandwich _____, Salad 8, Other 4. Sandwiches were chosen by 40% of the students.

- How many students are in the class?
- How many chose a sandwich?
- Find the relative frequency for Pizza.
- Name two appropriate displays for these data.

Answer Key and Explanations

1. Prospective study and variables

Answer

- Population: all tenth-grade students in the school district. Sample: the 800 selected tenth-grade students.
- Regular energy-drink use (yes/no) is categorical; average nightly sleep in hours is quantitative; final exam score is quantitative.
- Prospective, because the students were selected at the beginning and researchers then recorded a future outcome at the end of the semester.
- Possible answers: energy-drink use and sleep are self-reported; the sample may not represent the district; other factors such as workload or health may affect exam scores. This is observational, so confounding prevents a cause-and-effect conclusion.

2. Past records

Answer

- Retrospective, because archived records from past graduates were used.
- Whether the student took an art course is categorical; graduation GPA is quantitative.

3. Census or sample?

Answer

C. The population in context is all players on the current varsity team, and every one of those players was included.

4. Writing variables precisely

Answer

Many answers work. Examples: daily phone use in hours (quantitative); primary method of transportation to school (categorical).

5. One categorical variable

Answer

- Individuals: the 130 club members. Variable: primary way each member travels to school; categorical.
- Bus: $38/130 = 0.292 = 29.2\%$. Bicycle: $10/130 = 0.0769 = 7.7\%$.
- A bar graph or a pie chart.

6. Two-way table mastery

Answer

- a. Marginal: $58/240 = 0.2417 = 24.2\%$.
- b. Joint: $30/240 = 0.125 = 12.5\%$.
- c. Conditional on Senior: $30/80 = 0.375 = 37.5\%$.
- d. Conditional on Job: $30/64 = 0.46875 = 46.9\%$.

Notice that parts (c) and (d) use the same cell but different denominators.

7. Association in context

Answer

- a. Sports: Sophomores $36/80 = 45\%$; Juniors $30/80 = 37.5\%$; Seniors $24/80 = 30\%$.
- b. Job: Sophomores $12/80 = 15\%$; Juniors $22/80 = 27.5\%$; Seniors $30/80 = 37.5\%$.
- c. Yes. The conditional distribution of activity changes with grade level. For example, the percentage reporting a Job rises from 15% of Sophomores to 37.5% of Seniors, while the percentage reporting Sports falls from 45% to 30%.

8. Misleading display

Answer

Starting the axis at 92% makes small differences look much larger than they are. Redraw the frequency/percentage axis from 0% with equal intervals.

9. Mosaic plot without the picture

Answer

- App flashcards: High $20/40 = 50\%$, Medium $12/40 = 30\%$, Low $8/40 = 20\%$. Paper notes: High $24/80 = 30\%$, Medium $40/80 = 50\%$, Low $16/80 = 20\%$.
- The Paper notes bar is twice as wide because its group has 80 students compared with 40 in App flashcards.
- $20/120 = 0.1667 = 16.7\%$. The rectangle's area represents the joint proportion of all students who used App flashcards and earned a High score.
- Yes. The conditional score distributions differ: 50% of App flashcard students earned a High score compared with 30% of Paper notes students. This is association, not proof of causation.

10. Counts from a segmented bar graph

Answer

- No. Percentages alone do not reveal counts when group sizes are unknown.
- Group A: $(0.35)(80) = 28$. Group B: $(0.25)(160) = 40$. Group B has the larger count even though its percentage is lower.

11. No association

Answer

- High school must also be 60% Math and 40% English.

b.

	Math	English	Total
Middle school	60	40	100
High school	120	80	200
Total	180	120	300

- The bars would have identical segment heights: 60% Math and 40% English. The high school bar could contain more people, but a segmented bar graph normally gives both groups equal-width 100% bars.

12. Multiple choice

Answer

A. The widths show that School A has three times the group size. Because both groups have the same 40% “Yes” rate, School A must also have three times as many “Yes” responses: $0.40(3n) = 3 \cdot 0.40(n)$. (B) is false — the rates are equal. (D) is false for the same reason as (A) is true. (C) cannot be settled: matching “Yes” percentages say nothing about how the other 60% is split among the remaining response categories, so an association may still exist elsewhere in the distribution.

13. A numerator that spans two columns**Answer**

B. “For restaurants in City B” restricts you to the City B row, so the denominator is the row total 160, not the grand total 400. The numerator adds both sit-down columns: $\frac{50+20}{160} = \frac{70}{160} = 0.4375 = 43.75\%$. Choice (A) uses the wrong denominator; (C) invents a total; (D) answers a different question (the marginal share of City B).

14. Reading a segmented bar graph**Answer**

B. Within School Q’s bar both segments share the same denominator, so $70\% > 30\%$ is a valid comparison. (A) and (C) compare *counts* across bars, which a segmented bar graph hides because the bars are forced to equal width. (D) is false: School Q’s distribution differs from P’s and R’s, so there is an association.

15. Mosaic plot with counts**Answer**

- Period 3, because bar width is proportional to group size and Period 3 has the most students (35 of the 80 total).
- $(0.60)(20) + (0.40)(25) + (0.80)(35) = 12 + 10 + 28 = 50$ students.
- Yes. The passing rate varies from 40% in Period 2 to 80% in Period 3, so knowing the period helps predict whether a student passed.
- $28/80 = 0.35 = 35\%$ of all students. The rectangle’s area is the joint relative frequency: students who are in Period 3 *and* passed.

16. Running relative frequency backwards**Answer**

- The three known categories are $12 + 8 + 4 = 24$ students, which is the other 60%. So the total is $24/0.60 = 40$ students.
- $(0.40)(40) = 16$ students.
- $12/40 = 0.30 = 30\%$.
- A bar graph or a pie chart (one categorical variable).

Final self-check

Can you identify the denominator before touching the calculator? Can you compare conditional percentages in context? If yes, you are ready.